# LAPORAN TUGAS PEMROGRAMAN MOBILE

Disusun untuk Memenuhi Tugas Mata Kuliah

## Pemrograman Mobile

Dosen Pengampu:

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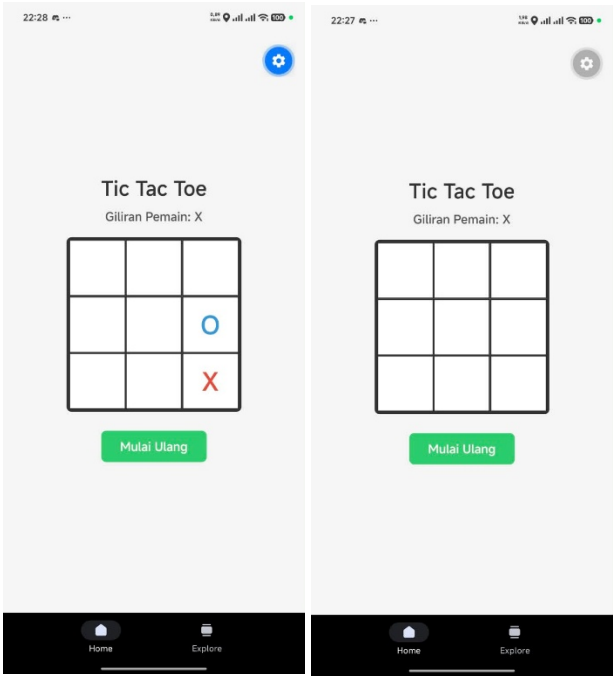
**FAKULTAS TEKNIK**

**PROGRAM STUDI INFORMATIKA**

**UNIVERSITAS LANGLANGBUANA**


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1. Screenshot papan game di HP saat ada pemenang, dan satu saat giliran berjalan.




2. Screenshot **terminal** yang menampilkan QR code.



3. Soucecode **index.tsx** di VSCode.

```
PS C:\Users\Jaffna Naufal\Downloads\senaart> cd
C:\Proyek\TicTacToeApp

PS C:\Proyek\TicTacToeApp> $target = if (Test-Path
"src\app\index.tsx") { "src\app\index.tsx" } else {
"app\index.tsx" }

>> $kode = @'

>> import React, { useState } from 'react';

>> import { StyleSheet, Text, View, TouchableOpacity } from
'react-native';

>> import { SafeAreaView } from 'react-native-safe-area-
context';

>>

>> type Square = 'X' | 'O' | null;

>>

>> export default function App() {

>>   /**

>>    * 1. STATE MANAGEMENT

>>    * - board: array 9 elemen (null, 'X', atau 'O')

>>    * - xIsNext: true untuk giliran 'X', false untuk 'O'

>>    */

>>   const [board, setBoard] =
useState<Square[]>(Array(9).fill(null));

>>   const [xIsNext, setXIsNext] = useState<boolean>(true);

>>

>>   /**

>>    * 2. LOGIKA PERMAINAN

>>    * Mendeteksi apakah ada pemain yang membentuk 3
simbol sejajar.

>>    */

>>   const calculateWinner = (squares: Square[]): Square =>
{

>>     const lines = [
```

```
>>       [0, 1, 2], [3, 4, 5], [6, 7, 8], // Baris horizontal
>>       [0, 3, 6], [1, 4, 7], [2, 5, 8], // Kolom vertikal
>>       [0, 4, 8], [2, 4, 6],            // Diagonal
>>     ];
>>     for (let i = 0; i < lines.length; i++) {
>>       const [a, b, c] = lines[i];
>>       if (squares[a] && squares[a] === squares[b] && squares[a] === squares[c]) {
>>         return squares[a]; // 'X' atau 'O' sebagai pemenang
>>       }
>>     }
>>     return null;
>>   };
>>
>>   const winner = calculateWinner(board);
>>   const isBoardFull = board.every((square) => square !== null);
>>
>>   /**
>>    * 3. EVENT HANDLER
>>    * Dijalankan saat kotak ditekan.
>>    */
>>   const handlePress = (index: number) => {
>>     // Abaikan klik jika kotak sudah terisi atau sudah ada pemenang
>>     if (board[index] || winner) return;
>>
>>     // IMMUTABILITY: buat salinan baru, jangan ubah state langsung
>>     const newBoard = [...board];
```

```
>>     newBoard[index] = xIsNext ? 'X' : 'O';
>>
>>     setBoard(newBoard);
>>     setXIsNext(!xIsNext);
>>   };
>>
>>   // Reset permainan ke kondisi awal
>>   const resetGame = () => {
>>     setBoard(Array(9).fill(null));
>>     setXIsNext(true);
>>   };
>>
>>   /**
>>    * 4. KOMPONEN KOTAK
>>    */
>>   const renderSquare = (index: number) => {
>>     return (
>>       <TouchableOpacity
>>         style={styles.square}
>>         onPress={() => handlePress(index)}
>>         activeOpacity={0.7}
>>       >
>>         <Text
>>           style={[
>>             styles.squareText,
>>             board[index] === 'X' ? styles.textX :
styles.textO,
>>           ]}
>>         >
>>           {board[index]}
>>         </Text>
```

```
>>       </TouchableOpacity>
>>     );
>>   };
>>
>>   // Teks status permainan
>>   let status;
>>   if (winner) {
>>     status = `Pemenang: ${winner}!`;
>>   } else if (isBoardFull) {
>>     status = 'Permainan Seri (Draw)!';
>>   } else {
>>     status = `Giliran Pemain: ${xIsNext ? 'X' : 'O'}`;
>>   }
>>
>>   return (
>>     <SafeAreaView style={styles.container}>
>>       {/* Judul Aplikasi */}
>>       <Text style={styles.title}>Tic Tac Toe</Text>
>>
>>       {/* Status Game */}
>>       <Text style={styles.status}>{status}</Text>
>>
>>       {/* Papan Permainan (Grid 3x3 dengan Flexbox) */}
>>       <View style={styles.board}>
>>         <View style={styles.row}>
>>           {renderSquare(0)}
>>           {renderSquare(1)}
>>           {renderSquare(2)}
>>         </View>
>>         <View style={styles.row}>
```

```
>>           {renderSquare(3)}
>>           {renderSquare(4)}
>>           {renderSquare(5)}
>>         </View>
>>         <View style={styles.row}>
>>           {renderSquare(6)}
>>           {renderSquare(7)}
>>           {renderSquare(8)}
>>         </View>
>>       </View>
>>
>>       {/* Tombol Reset */}
>>       <TouchableOpacity style={styles.resetButton}
onPress={resetGame}>
>>         <Text style={styles.resetButtonText}>Mulai
Ulang</Text>
>>       </TouchableOpacity>
>>     </SafeAreaView>
>>   );
>> }
>>
>> /**
>>  * 5. STYLING
>>  */
>> const styles = StyleSheet.create({
>>   container: {
>>     flex: 1,
>>     backgroundColor: '#f5f5f5',
>>     alignItems: 'center',
>>     justifyContent: 'center',
>>     padding: 20,
```

```
>>   },
>>   title: {
>>     fontSize: 32,
>>     fontWeight: 'bold',
>>     marginBottom: 10,
>>     color: '#333',
>>   },
>>   status: {
>>     fontSize: 18,
>>     marginBottom: 20,
>>     fontWeight: '600',
>>     color: '#555',
>>   },
>>   board: {
>>     borderWidth: 4,
>>     borderColor: '#333',
>>     borderRadius: 8,
>>     backgroundColor: '#fff',
>>   },
>>   row: {
>>     flexDirection: 'row', // Kotak berderet horizontal
>>   },
>>   square: {
>>     width: 90,
>>     height: 90,
>>     borderWidth: 2,
>>     borderColor: '#333',
>>     alignItems: 'center',
>>     justifyContent: 'center',
>>   },
```

```
>>   squareText: {
>>     fontSize: 40,
>>     fontWeight: 'bold',
>>   },
>>   textX: {
>>     color: '#e74c3c', // Merah untuk X
>>   },
>>   textO: {
>>     color: '#3498db', // Biru untuk O
>>   },
>>   resetButton: {
>>     marginTop: 30,
>>     backgroundColor: '#2ecc71',
>>     paddingHorizontal: 30,
>>     paddingVertical: 12,
>>     borderRadius: 8,
>>   },
>>   resetButtonText: {
>>     color: '#fff',
>>     fontSize: 18,
>>     fontWeight: 'bold',
>>   },
>> });
>> '@
>> Set-Content -Path $target -Value $kode -Encoding UTF8
>> Write-Host "Kode ditulis ke: $target"
Kode ditulis ke: src\app\index.tsx
```